Models of computation

Advanced Data Structures – Summer 2026 – Lecture 1

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Today

- Why do we care about models?
- Big-O notation
- The complexity of sorting
- Properties data structures
- The word RAM model
  - Variations
  - Limitations

Motivation

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We want to design fast algorithms, but when is an algorithm *fast*?

- Implement it, and *measure* the time.
  - Extrapolate experiments to predict performance on larger n.
  - May or may not give understanding.
  - Does not provide lower bounds.
- Analyse exactly what happens on the CPU.
  - Unwieldy; they are massive black boxes.
- Instead: use a simplified abstract *model* of the CPU.
  - Allows exact lower and upper bounds.

Big-O notation (recap)

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Big-O, or $\mathcal{O}$ notation relates the asymptotic growth of two abstract functions to each other.

Definition 1 Big-O.

Given two functions $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$, we say that

$$f(n) = O(g(n)), \text{ or}$$
$$f(n) \in O(g(n))$$

when there exists constants $n_0 \in \mathbb{N}$ and $M \in \mathbb{R}$ such that for all sufficiently large $n \geq n_0$:

$$f(n) \leq M \cdot g(n).$$

Example 1.

$$n = O(n) \qquad \ln n = O(\sqrt{n})$$
$$n = O(n^2) \qquad (\lg n)^{100} \cdot n = O(n^{1.0001})$$
$$1/n = O(1) \qquad \ln n! = n \lg n - n + O(\lg n) = O(n \lg n)$$

Definition 2 Extended notation.

- $f(n) = o(g(n))$ is like $<$:

$$\lim_{n\to\infty} f(n)/g(n) \to 0.$$

- $f(n) = O(g(n))$ is like $\leq$:

$$f(n) \leq M \cdot g(n) \quad \forall n \geq n_0.$$

- $f(n) = \Theta(g(n))$ is like $=$:

$$f(n) = O(g(n)) \text{ and } g(n) = O(f(n)).$$

- $f(n) \sim g(n)$ is a stronger Θ:

$$\lim_{n\to\infty} f(n)/g(n) = 1.$$

- $f(n) = \Omega(g(n))$ is like $\geq$:

$$f(n) \geq M \cdot g(n) \quad \forall n \geq n_0.$$

- $f(n) = \omega(g(n))$ is like $>$:

$$\limsup_{n\to\infty} f(n)/g(n) \to \infty.$$

Quiz time!

What is the complexity of sorting?

What is the complexity of sorting?

- $O(n \lg n)$?
- $O(n \lg w)$?
- $O(n\sqrt{\lg n})$?
- $O(n\sqrt{\lg \lg n})$?
- Linear $O(n)$?
- Constant $O(1)$?
- Something else?

What is the complexity of sorting?

All of the above!

- $O(n \lg n)$
  - Minimum number of *comparisons* is $\Omega(n \lg n)$.
  - Merge sort indeed uses $O(n \lg n)$ of them.
  - Or QuickSort, with $O(n \lg n)$ comparisons *in expectation*.
- $O(n \lg w)$
  - Expected complexity of Van Emde-Boas trees (van Emde Boas 1975).
- $O(n\sqrt{\lg n})$
  - Fusion trees (Fredman and Willard 1993)
- $O(n\sqrt{\lg \lg n})$, $O(n\sqrt{\lg w})$
  - Fastest *randomized* algorithm (Han and Thorup, n.d.)
- Linear $O(n)$
  - Linear *space*? Sure!
  - Counting sort when all values are $O(n)$.
  - Radix sort on 64-bit integers.
  - Randomized when $w \geq (\lg n)^{2+\varepsilon}$ (Andersson et al. 1995)
- Constant $O(1)$
  - $n \leq 2^{64}$ makes $f(n)$ bounded.
  - There's only so many atoms in the universe.

ALGORITHM 64
QUICKSORT
C. A. R. HOARE
Elliott Brothers Ltd., Borehamwood, Hertfordshire, Eng.

```
procedure  quicksort (A,M,N);  value M,N;
             array A;  integer M,N;
comment  Quicksort is a very fast and convenient method of
sorting an array in the random-access store of a computer. The
entire contents of the store may be sorted, since no extra space is
required. The average number of comparisons made is 2(M−N) ln
(N−M), and the average number of exchanges is one sixth this
amount. Suitable refinements of this method will be desirable for
its implementation on any actual computer;
begin      integer I,J;
           if M < N then begin partition (A,M,N,I,J);
                                quicksort (A,M,J);
                                quicksort (A, I, N)
                          end
end        quicksort
```

Figure 1: Algorithm 64: Quicksort (Hoare 1961). RIP Tony Hoare.

"Complexity"

"Complexity"

- Lower or upper bound?
  - Complexity of a *problem* is at least $\Omega(\ldots)$.
  - Complexity of an *algorithm* at most $O(\ldots)$.
  - "The complexity": matching lower and upper bound.
- Which metric?
  - "Operations"
  - Comparisons
  - Memory accesses
  - Memory touched
  - Wall time
  - Space usage
- What input type?
  - Integers
  - Floats
  - Abstract "*objects*"
- What input properties?
  - (Uniform?) random
  - Worst-case
- What algorithmic properties?
  - Deterministic
  - Randomized

Properties of data structures

Space usage

Let S be the minimum space required to represent some data.

Definition 3 Compact data structure.

A *compact* data structure uses $O(S)$ bits of space.

Definition 4 Succinct data structure.

A *succinct* data structure uses $S + o(S)$ bits of space.

Example: many Rank & Select data structures are succinct.

Definition 5 Implicit data structure.

An *implicit* data structure uses $S + O(1)$ bits of space.

Example: bit arrays.

Dynamic data structures

Definition 6 Dynamic data structure.

Allows mutating the structure.

Example: hash tables.

Definition 7 Static data structure.

Build once, then read-only queries.

Example: minimal perfect hash functions.

Definition 8 Incremental data structure.

Insert-only; deletions not allowed.

Example: bloom filters.

Running time

From stronger to weaker:

Definition 9 Worst-case running time.

An algorithm has *worst-case* running time $O(f(n))$ when for *every possible* input, it is guaranteed to finishes in $O(f(n))$ time.

Definition 10 Expected running time.

An algorithm has *expected* running time $O(f(n))$ when for *every possible* input, the *expected* running time is $O(f(n))$.

Definition 11 Running time with high probability.

An algorithm has running time $O(f(n))$ *with high probability* (w.h.p.) when for *every possible* input, the running time is $O(f(n))$ with probability $1 - o(1)$.

Running time

Definition 12 Amortized running time.

An operation on a data structure has *amortized* running time $O(f(n))$ when for *every possible* sequence of i operations with amortized complexities f_i, the total running time is $O(\sum_i f_i)$.

Example: pushing on a vector is worst-case $O(n)$, but amortized $O(1)$.

Models of computation

The RAM model

Definition 13 Random-access machine (RAM) model.

Models a machine that has an *infinite* list of *registers*.

- Each register can store an **unbounded (!)** natural number!
- Each register has an *address*.
- Retrieving the register corresponding to an address takes constant time.

Question 1.

Any problems with this?

The *real* RAM model

Definition 14.

Like the RAM model, but each register can store a *real* number.

The *word* RAM model

Definition 15.

Like the RAM model, but operates on *words* of w bits.

- Registers are also w bits.
- Constant-time w-bit arithmetic operations:
  - bit-operations, addition, multiplication.

Evolved with CPUs to include: https://en.wikipedia.org/wiki/X86_Bit_manipulation_instruction_set

- Popcount (`popcnt`), count trailing zeros (`tzcnt`, BMI1)
- Parallel bit deposit (`pdep`), parallel bit extract (`pext`, BMI2)

Most commonly used model.

Question 2.

Any remaining problems with *this*?

We must talk about w

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- In practice, $w = 64$ is constant.
- But the problem size n grows to infinity.
- What if $n > 2^w$? Then we can't even represent n?

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This has far-reaching implications!

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Solution: we assume

$$\lg n \leq w.$$

This has far-reaching implications!

- The complexity of pop-counting n bits is **sub-linear** $O(n/\lg n)$!
- Fusion trees can binary search on $O(\sqrt{w})$ w-bit integers in *constant* $O(1)$ time!

The pointer model

Definition 16 Pointer model.

RAM-model without the RAM:

- Registers can only be accessed via direct pointers.

Used in the analysis of some heap-based priority queues.

The cell-probe model

Definition 17 Cell-probe model.

Like the RAM-model, but all operations apart from accessing memory are free.

- Typically, an infinite cache is assumed, so that each address only has to be read once.
- Used for lower bounds on the number of addresses that must be read to solve some task, and thus a lower bound on RAM-model complexity.

The RAM-model is unphysical

The RAM-model is unphysical

- Infinite memory with constant time access is fundamentally not possible:
  - Each bit takes some small volume v to store.
  - Signals can only travel at the speed of light c.
  - In time t, we can read from a sphere of radius tc, containing

$$\frac{4}{3}\pi(tc)^3/v = \Theta(t^3) \quad \text{bits.}$$

Observation 1 Latency.

The *latency* of a uniform random access into a memory of size n is at least

$$\Omega(\sqrt[3]{n}).$$

The RAM-model is unphysical (2)

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- Black holes have only *quadratic* mass $\Theta(r^2)$!
  - Filling a volume with cubic information is *impossible*.
  - But we are far away from the constant.
- If each bit of memory requires some energy, we need to dissipate $\Theta(r^3)$ heat through $\Theta(r^2)$ of surface area.
  - This is an actual bottleneck for CPUs! They are mostly flat (2.5D) for a reason!
    - AMD now has "3D V-Cache" which vertically stacks caches.

Observation 2 Black holes and cooling (handwavy).

We can store/cool at most $O(r^2)$ bits in a sphere of radius r, and so the *latency* of a uniform random access into a memory of size n is at least

$$\Omega(\sqrt[2]{n}).$$

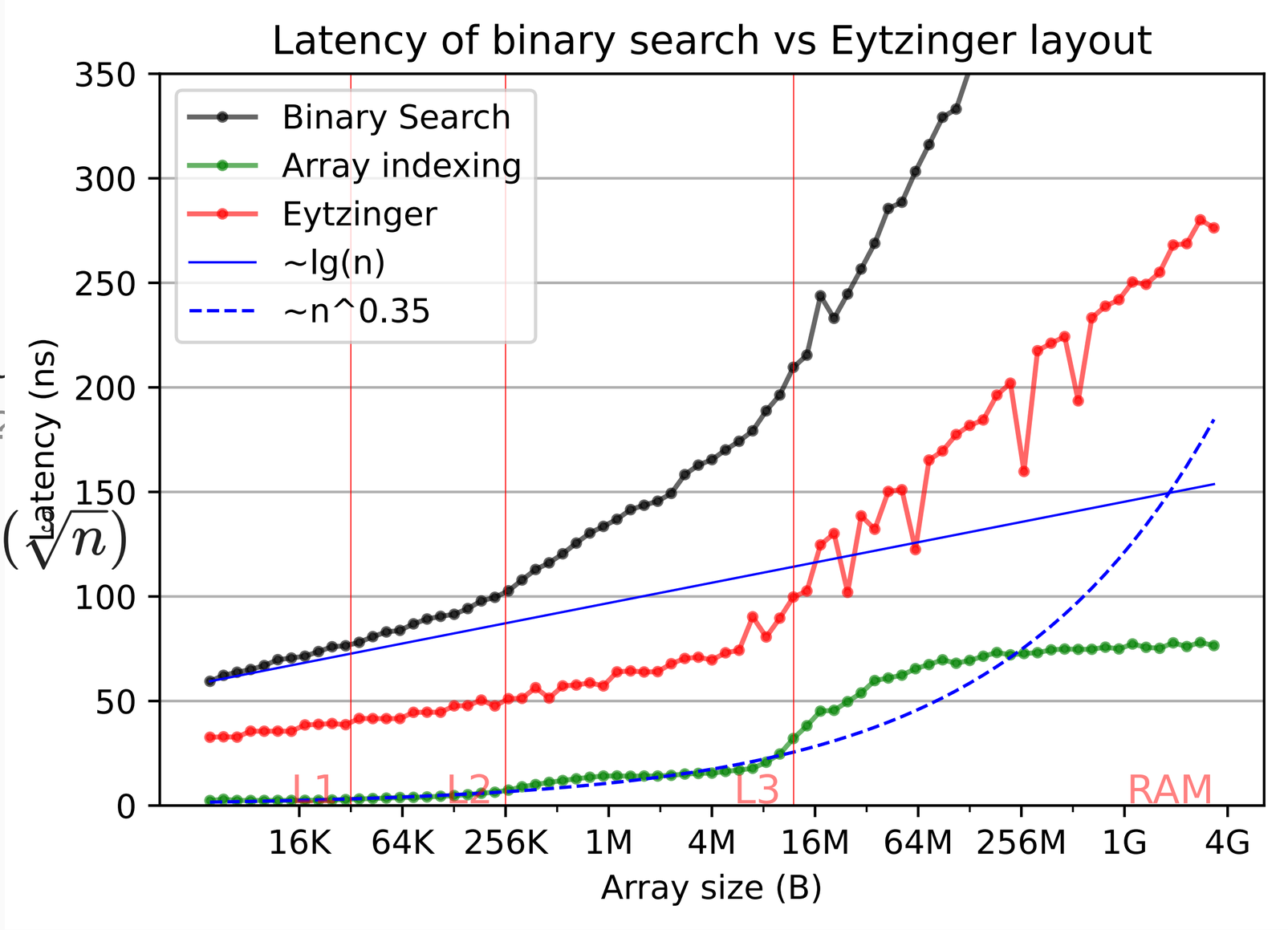
- See this blog: The Myth of RAM

The RAM-model is unrealistic

- Green: latency of array indexing.
  - Clearly not $O(1)$!
  - More like $\sqrt[3]{n}$ (dashed blue)
- Black: latency of binary search.
  - Clearly not $O(\lg n)$ (solid blue)
- Red: binary search with heap-layou
  - $< 3\times$ slower than just indexing?

$$\sqrt[3]{1} + \sqrt[3]{2} + \cdots + \sqrt[3]{n} = \Theta(\sqrt[3]{n})$$

- **TODO:** $\sqrt{n}$-complexity model!



The I/O-complexity

Introduced by Aggarwal and Vitter (1988).

Definition 18 I/O-complexity.

Like the cell-probe model, the I/O-complexity only counts I/O operations.

- Fast and free *internal* memory of M words.
- Slow (infinite) *external* memory.
- Count I/O-operations between them go in *blocks* of B words.

Theorem 1 I/O-complexity of sorting.

Sorting n word-sized integers has an I/O-complexity lower-bound of

$$\Omega\left(\frac{n}{B}\log_{M/B}\frac{n}{B}\right),$$

and external merge-sort achieves this.

Takeaways

Observation 3.

We need precisely defined models to theoretically analyse algorithms.

Observation 4.

Models approximate reality; some do so better than others.

The rest of this course will mostly use the **word RAM model** to analyse both theoretically and practically efficient algorithms.

Further reading

- These slides: curiouscoding.nl/teaching
- Algorithmica on binary search:
  - https://en.algorithmica.org/hpc/data-structures/binary-search/
  - Great resource for understanding and optimizing for modern CPUs
- Paper: Array layouts for comparison-based searching (Khuong and Morin 2017)
- Blog: The Myth of RAM
  - https://www.ilikebigbits.com/2014_04_21_myth_of_ram_1.html
- My P99 talk on binary search
  - https://curiouscoding.nl/slides/p99/
  - recording on youtube
- Aggarwal and Vitter's slides on I/O-complexity:
  - https://jshun.csail.mit.edu/6886-s19/lectures/lecture5-1.pdf

Next week: Bitvectors, Rank, & Select

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Possible exam questions

- What is big-O notation? Name some of the different variants.
- What are some problems with saying "the complexity of sorting"?
- Give an example of a precisely specified complexity of an algorithm.
- What is the difference between compact, succinct, and implicit data structures? Give an example for each of them.
- What is the difference between worst-case, expected, and amortized running time?
- Name 5 different models of computation.
- What are some problems with the word RAM model? Why do we use it anyway? Is there a model that solves these issues?
- What can you say about w?